WEEK 11

AUTOMOBILE DATA CLEANING & TRANSFORMATION

Introduction :

The Automobile Dataset contains information about different vehicles, including their engine specifications, fuel efficiency, body style and price.

The original dataset contains missing values and invalid symbols. So, data cleaning and transformation are important before performing analysis.

In this project, I cleaned the automobile data, handled missing values, created new calculated columns, grouped vehicles into different categories and created DAX measures. These steps help to prepare the dataset for accurate analysis and comparison.

Objectives :

- Replace invalid symbols (?) with Null values.
- Change technical columns into correct data types.
- Handle missing values using conditional logic.
- Create new calculated columns.
- Group vehicles based on different categories.
- Create DAX measures for data analysis.
- Prepare a clean automobile data model.

Data Cleaning in Power Query :

Replace Invalid Values :

The original dataset contained the symbol (?) in some columns.

I used the Replace Values option in Power Query to replace (?) with Null values in the following columns:

1. Normalized Losses
2. Number of Doors
3. Bore

4. Stroke
5. Horsepower
6. Peak RPM
7. Price

This step helps to identify missing values and prevents invalid symbols from creating errors during calculations.

Change Data Types :

I changed the data types of the columns according to their values in Power Query.

1. Normalized Losses – Whole Number
2. Number of Doors – Text
3. Bore – Decimal Number
4. Stroke – Decimal Number
5. Horsepower – Whole Number
6. Peak RPM – Whole Number
7. Price – Whole Number
8. Curb Weight – Whole Number
9. Highway MPG – Whole Number

Correct data types help Power BI perform calculations properly and avoid errors during data transformation.

Missing Value Imputation :

Missing values were handled using Custom Columns in Power Query.

For numerical columns, I used the Median value to fill missing values. The Median is used because it helps to fill missing values without being affected too much by very high or very low values.

I created the following six new columns using Power Query:

Horsepower Filled :

This column fills missing horsepower values using the Median of the available horsepower values.

Formula:

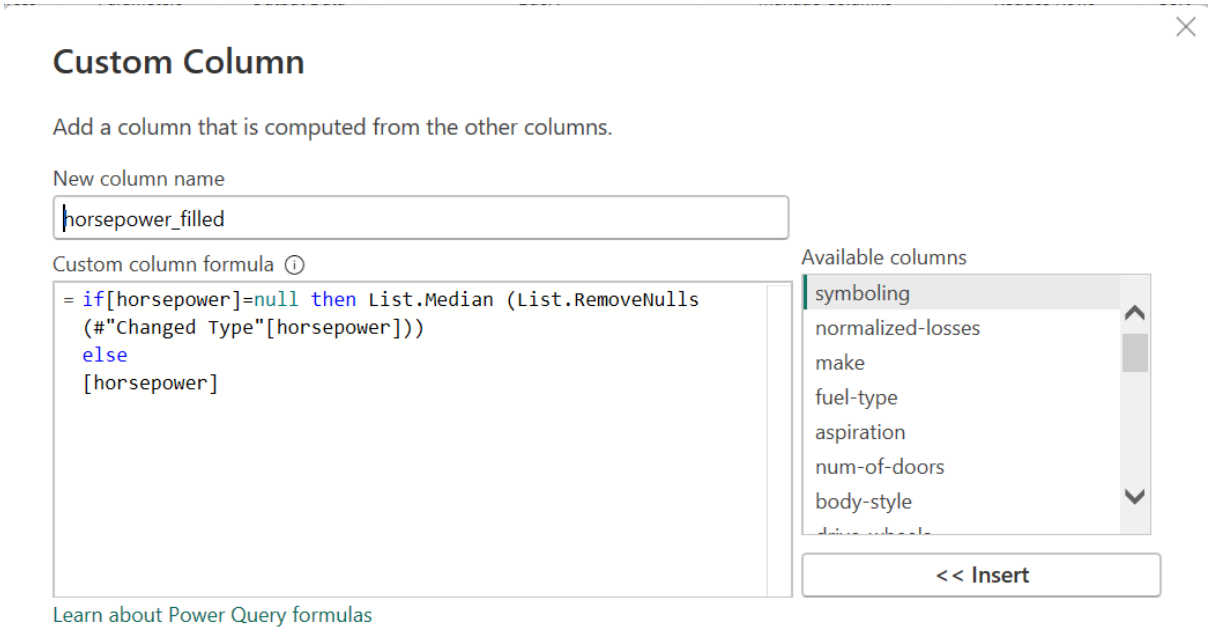
if [horsepower] = null then

List.Median(List.RemoveNulls(#"Changed Type"[horsepower]))

else

[horsepower]

New column: horsepower_filled



Peak RPM Filled :

This column fills missing peak RPM values using the Median of the available peak RPM values.

Formula:

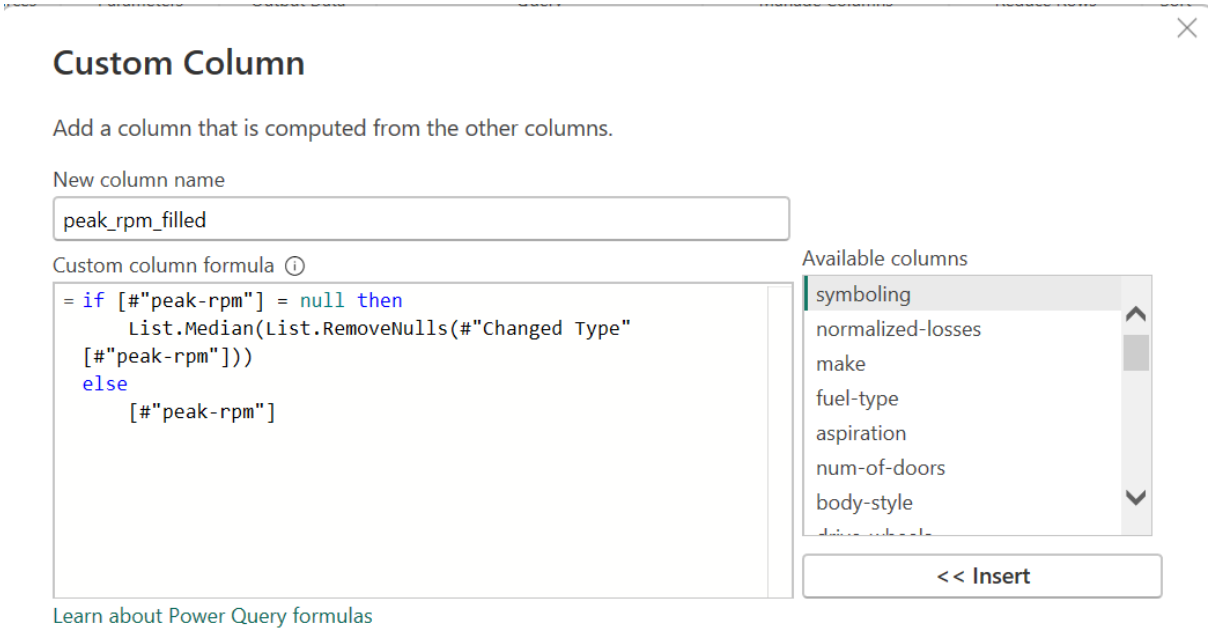
if [#"peak-rpm"] = null then

List.Median(List.RemoveNulls(#"Changed Type"[#"peak-rpm"]))

else

[#"peak-rpm"]

New column: peak_rpm_filled



Price Filled :

This column fills missing price values using the Median of the available price values.

Formula:

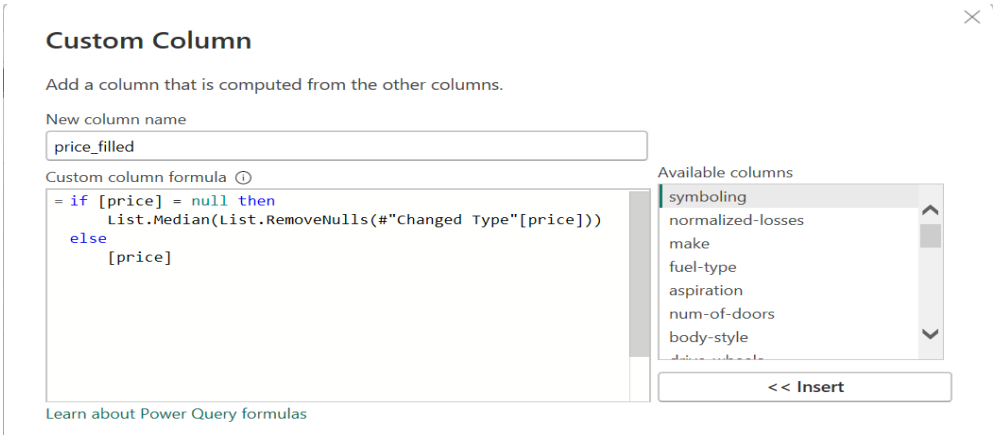
if [price] = null then

List.Median(List.RemoveNulls(#"Changed Type"[price]))

else

[price]

New column: price_filled



Normalized Losses Filled :

This column fills missing normalized losses values using the Median of the available normalized losses values.

Formula:

if [#"normalized-losses"] = null then

List.Median(List.RemoveNulls(#"Changed Type3"[#"normalized-losses"]))

else

[#"normalized-losses"]

New column: normalized_losses_filled



Bore Filled :

This column fills missing bore values using the Median of the available bore values.

Formula:

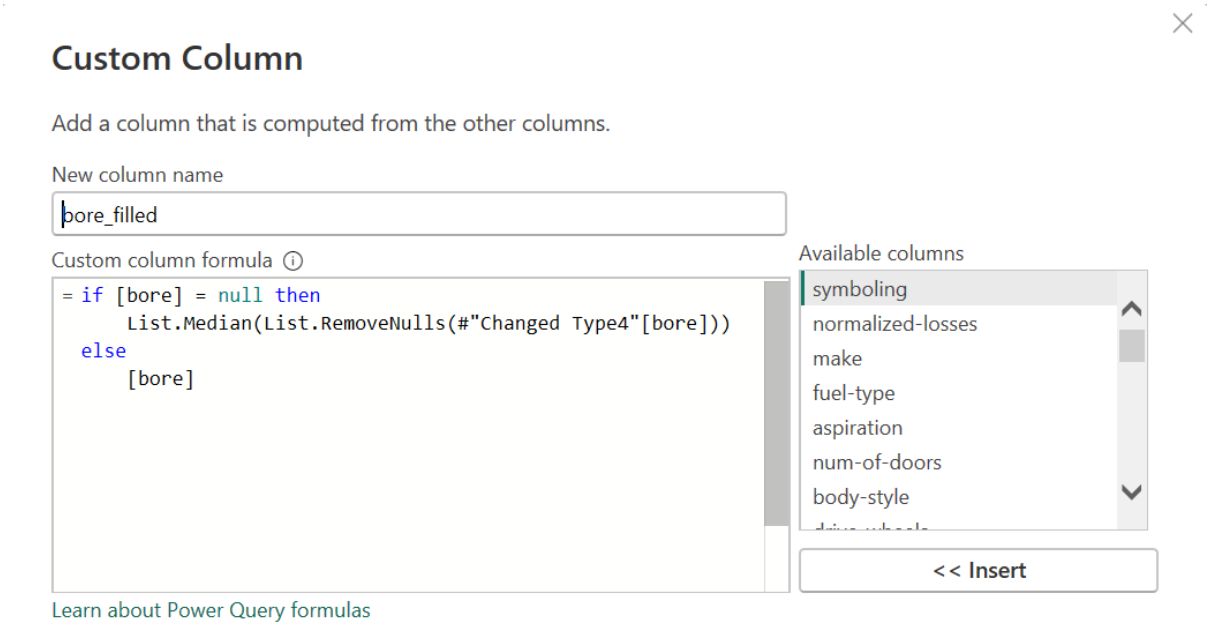
if [bore] = null then

List.Median(List.RemoveNulls(#"Changed Type4"[bore]))

else

[bore]

New column: bore_filled



Stroke Filled :

This column fills missing stroke values using the Median of the available stroke values.

Formula:

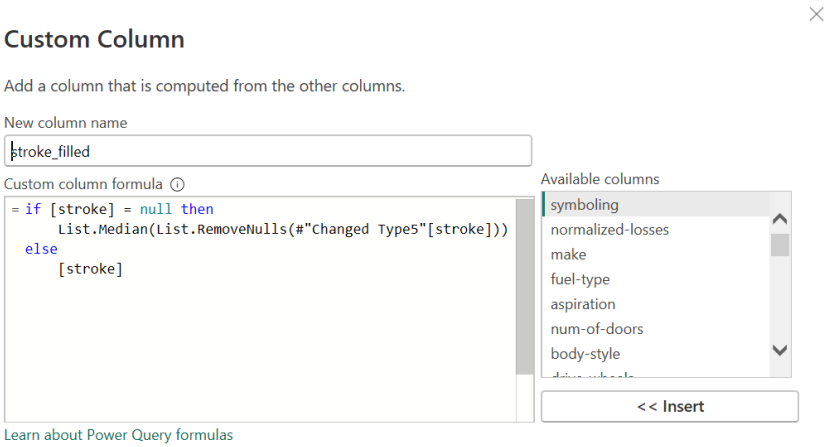
if [stroke] = null then

List.Median(List.RemoveNulls(#"Changed Type5"[stroke]))

else

[stroke]

New column: stroke_filled



Number of Doors Filled :

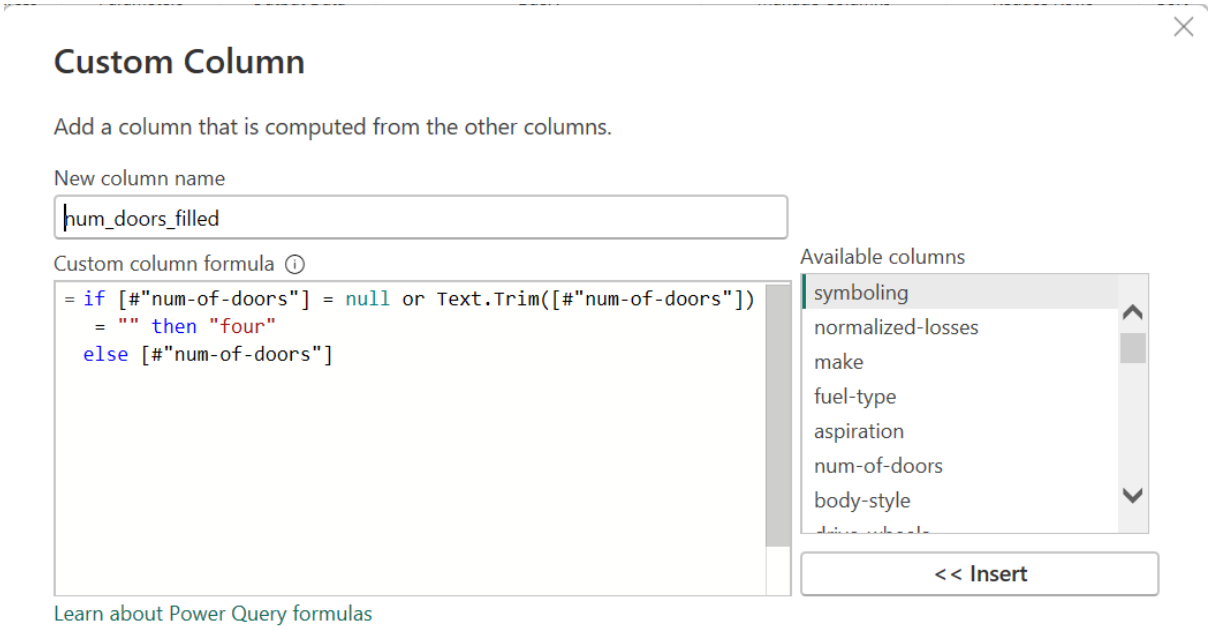
For the Number of Doors column, I used the value "four" to fill missing or empty values.

Formula:

if [#"num-of-doors"] = null or Text.Trim([#"num-of-doors"]) = "" then "four"

else [#"num-of-doors"]

New column: num_doors_filled



These steps help to reduce missing values in important columns and prepare the dataset for further calculations and analysis.

Derived Feature Creation :

I created three new calculated columns to compare vehicle performance, price, weight and fuel efficiency.

Price per Horsepower :

This column calculates the price of a vehicle for each unit of horsepower. It helps to compare vehicle prices based on engine power.

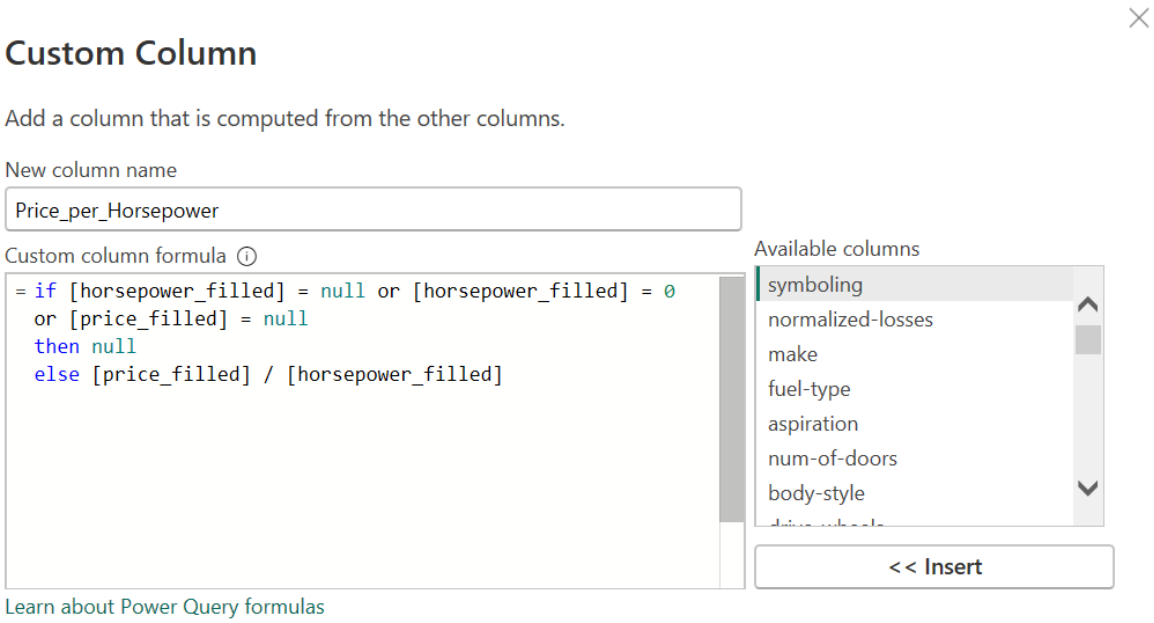
Formula:

if [horsepower_filled] = null or [horsepower_filled] = 0 or [price_filled] = null

then null

else [price_filled] / [horsepower_filled]

New column: Price_per_Horsepower



Power-to-Weight Ratio :

This column calculates the relationship between engine horsepower and vehicle weight. It helps to compare the power of different vehicles based on their weight.

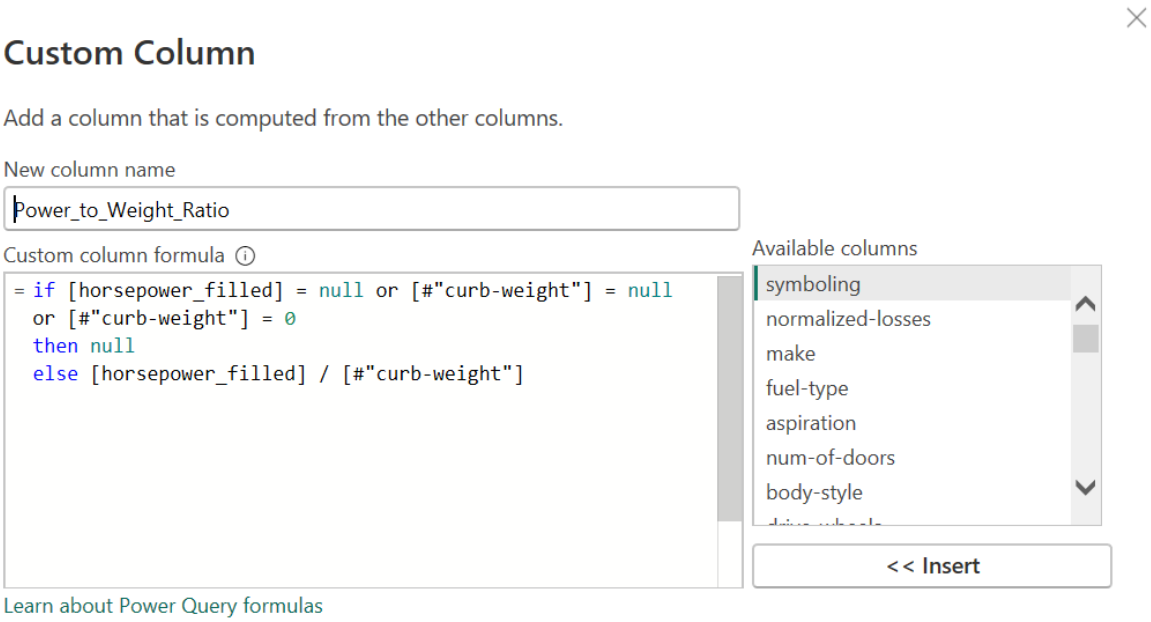
Formula:

if [horsepower_filled] = null or [#"curb-weight"] = null or [#"curb-weight"] = 0

then null

else [horsepower_filled] / [#"curb-weight"]

New column: Power_to_Weight_Ratio



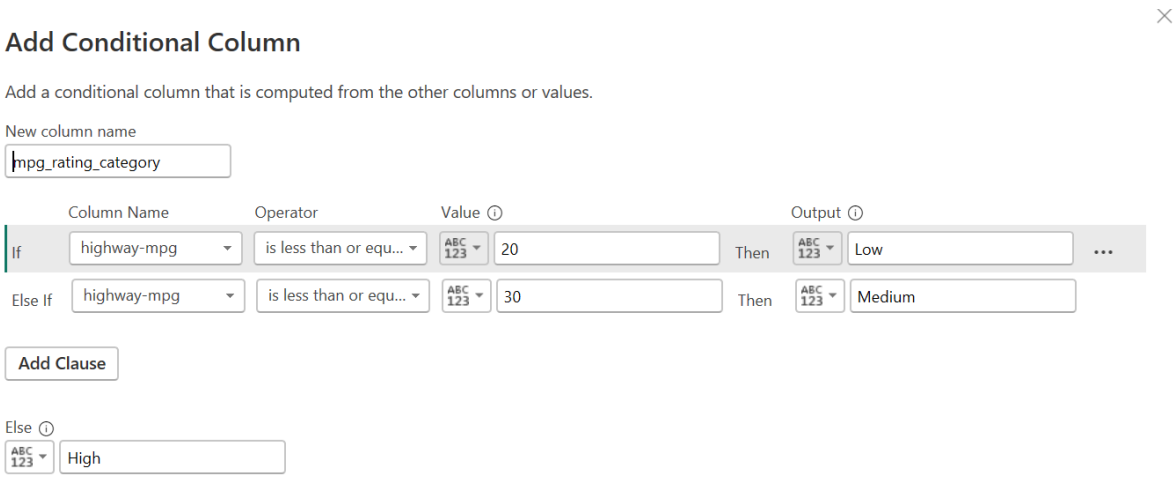
MPG Rating Category :

I used the Conditional Column option to classify vehicles based on Highway MPG. This makes it easier to identify vehicles according to their fuel efficiency.

Conditions:

1. Highway MPG 20 or below – Low
2. Highway MPG from 21 to 30 – Medium
3. Highway MPG above 30 – High

New column: MPG_Rating_Category



Grouping Vehicles :

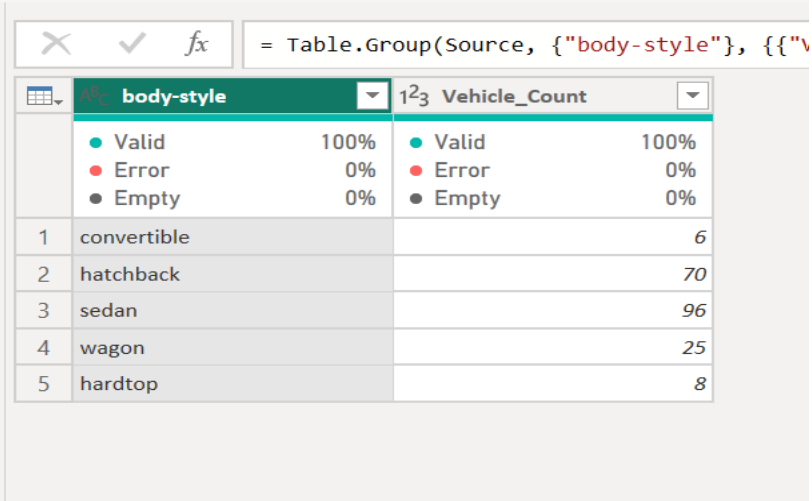
I used the Group By option in Power Query to group vehicles into different categories. The operation used was Count Rows, which calculates the number of vehicles in each category.

Body Style Grouping :

I grouped vehicles using the body-style column.

New column: Vehicle_Count

This grouping shows how many vehicles belong to each body style, such as sedan, hatchback and wagon.

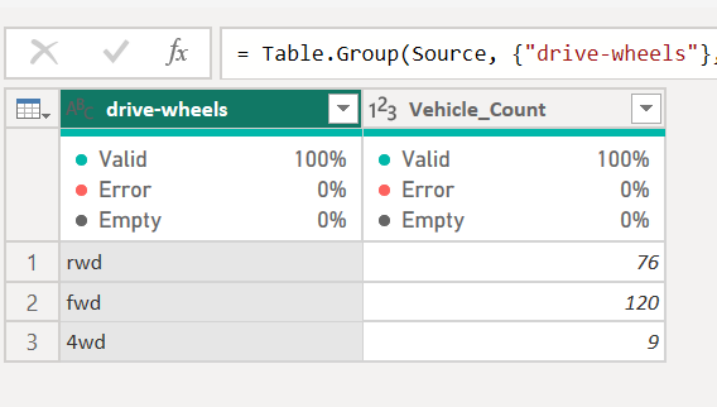


Drive Wheels Grouping :

I grouped vehicles using the drive-wheels column.

New column: Vehicle_Count

This grouping shows the number of vehicles based on their drive wheel types, such as FWD, RWD and 4WD.

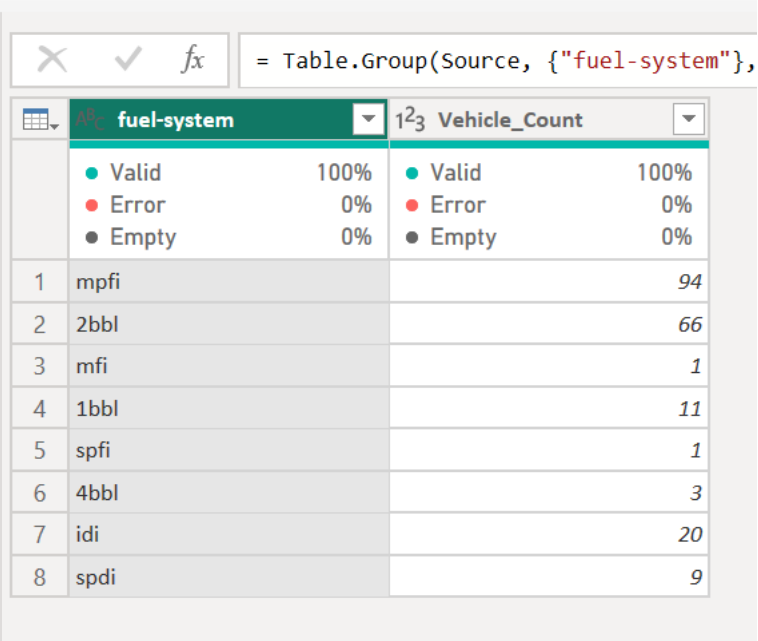


Fuel System Grouping :

I grouped vehicles using the fuel-system column.

New column: Vehicle_Count

This grouping shows how many vehicles use each type of fuel system.



	fuel-system	Vehicle_Count
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	mpfi	94
2	2bbl	66
3	mfi	1
4	1bbl	11
5	spfi	1
6	4bbl	3
7	idi	20
8	spdi	9

Data Model and Relationships :

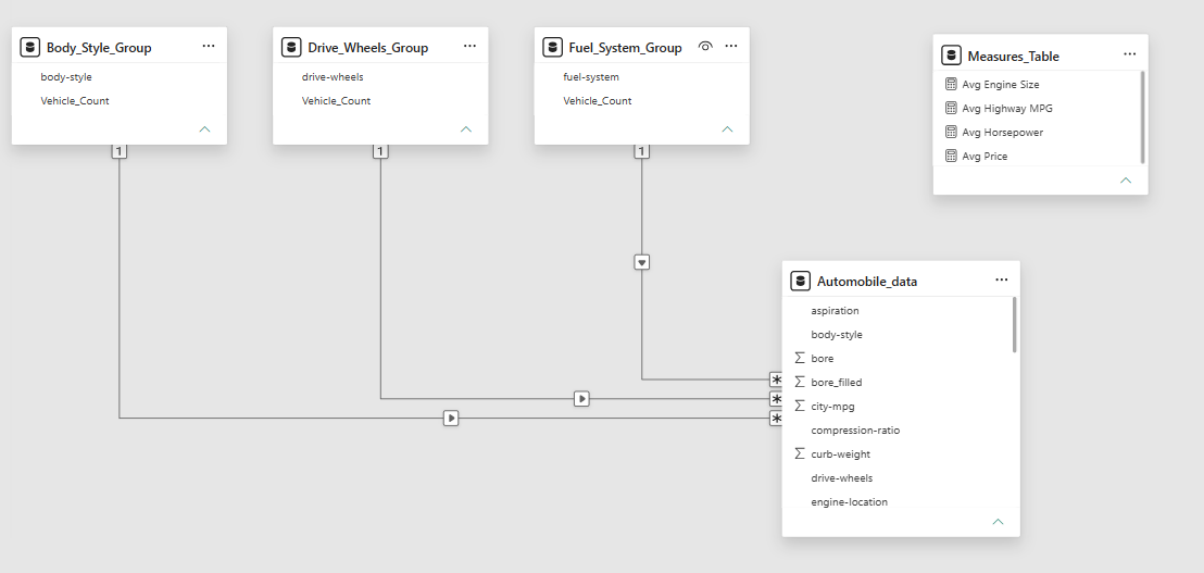
I created separate grouped queries using the Reference option in Power Query.

The grouped tables are:

1. Body_Style_Group
2. Drive_Wheels_Group
3. Fuel_System_Group

I checked the relationships in Model View. The grouped tables contain unique category values, while the original automobile table contains multiple vehicle records for each category.

Therefore, the grouped category tables are on the One (1) side and the original automobile table is on the Many (*) side. These relationships help connect category information with the original vehicle records.



DAX Measures :

I created four DAX measures to calculate the average values of important automobile features. These measures help to understand the average price, horsepower, engine size and highway fuel efficiency of the vehicles.

Average Price :

Avg Price = AVERAGE(Automobile_data[price_filled])

Calculates the average price of vehicles.

Average Horsepower :

Avg Horsepower = AVERAGE(Automobile_data[horsepower_filled])

Calculates the average horsepower of vehicles.

Average Engine Size :

Avg Engine Size = AVERAGE(Automobile_data[engine-size])

Calculates the average engine size of vehicles.

Average Highway MPG :

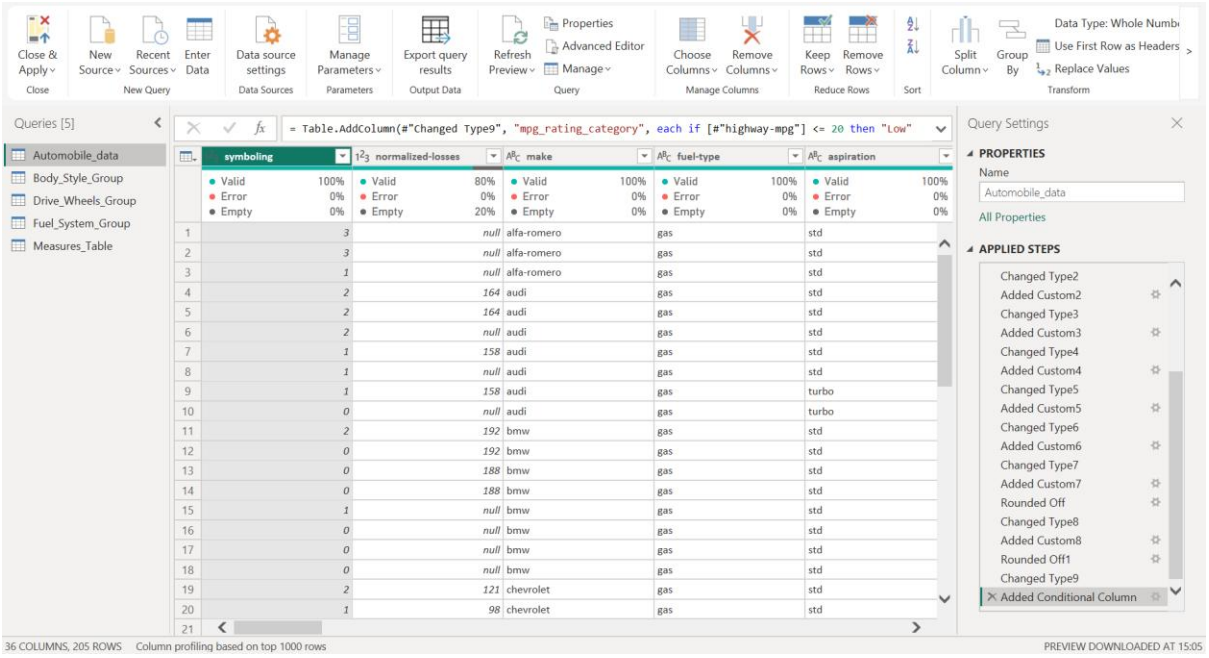
Avg Highway MPG = AVERAGE(Automobile_data[highway-mpg])

Calculates the average highway fuel efficiency of vehicles.

Final Data Model :

The automobile dataset was cleaned and transformed using Power Query. The final data contains cleaned automobile records, correct data types, filled missing values, new calculated feature columns, grouped vehicle categories and four DAX measures.

The cleaned data model is now ready for further automobile data analysis.



Conclusion :

In this project, I performed data cleaning and transformation on the Automobile Dataset using Power BI. I replaced invalid values, handled missing data, changed data types and created new calculated columns. I also grouped vehicles based on body style, drive wheels and fuel system. Finally, I created four DAX measures to calculate average price, horsepower, engine size and highway MPG. This project helped me understand how to clean, transform and prepare automobile data for analysis using Power BI.